

All-Optical Wavelength Conversion by Electroabsorption Modulator

Kohsuke Nishimura, Ryo Inohara, Masashi Usami, *Member, IEEE*, and Shigeyuki Akiba, *Fellow, IEEE*

Abstract—All-optical wavelength conversion using an electroabsorption modulator (EAM) in conjunction with delayed interferometer (DI) is investigated. The optical nonlinearities induced in an EAM were experimentally characterized. High bit-rate wavelength conversion at 100 Gb/s was demonstrated by an EAM-DI wavelength converter with required optical pulse energy of 1.5 pJ. It was verified that the operable bandwidth of the EAM-DI wavelength converter at 40 Gb/s covered almost full range of C-band without tuning operation conditions. These results indicate that the EAM-based wavelength converter is capable for ultrafast wide-band wavelength conversion.

Index Terms—All-optical wavelength conversion, delayed interferometer (DI), electroabsorption modulator (EAM).

I. INTRODUCTION

BROADBAND access services such as digital subscriber line and fiber-to-the-home are getting more and more popular, and the rapid increase of broadband access capacity will trigger the expansion of capacity of metro area network as well as backbone network. Most up-to-date wavelength division multiplexing (WDM) technologies realize optical link capacity exceeding 10 Tb/s per one fiber. To handle such a large capacity at a routing node in a cost-effective manner, all-electronic signal processing may not be realistic due to large power consumption resulting in large heat dissipation. The first step to involve all-optical processing in a routing node is the employment of optical matrix switch, which can be implemented by microelectromechanical system technology or planar lightwave circuit technology. The optical matrix switch can cut through the electronic process for the data traffics that are unnecessary to be dropped at the node. However, if we eliminate all the optical-to-electric-to-optical (OEO) conversion from a routing node, a different type of problem will be caused: i.e., contention among wavelength channels. If one output port for certain wavelength channel is occupied by preceding data traffic, other succeeding data traffic with the same wavelength channel cannot be routed to the output port until the preceding data traffic releases the port. All-optical wavelength conversion can solve such a problem of wavelength channel contention, and hence it is one of the key technologies to improve scalability of future all-photonics network.

Manuscript received August 2, 2004; revised October 26, 2004. A part of this work was performed in "Photonic Network Project" which OITDA contracted with New Energy and Industrial Technology Development Organization (NEDO).

K. Nishimura, R. Inohara and M. Usami are with KDDI R&D Laboratories Inc., Kamifukuoka, Saitama 356-8502, Japan. They are also with OITDA, Bunkyo-ku, Tokyo 112-0014, Japan (e-mail: nish@kddilabs.jp).

S. Akiba is with KDDI Corporation, Chiyoda-ku, Tokyo 102-8460, Japan.
Digital Object Identifier 10.1109/JSTQE.2004.841708

A variety of all-optical wavelength converters have been proposed and investigated so far. Various kinds of optical nonlinearity are possible to be utilized for all-optical wavelength conversion, although there is a significant problem of tradeoff between required optical power and response speed. For example, Kerr effect induced in silica fiber has subpicosecond response, but its efficiency is quite low and it requires several hundred milliwatt class optical pump power to induce phase shift large enough to be utilized for wavelength conversion [1]–[5]. On the other hand, a semiconductor optical amplifier (SOA) is superior in efficiency due to its gain, and wavelength conversion with a few or submilliwatt class pump power is possible [6]–[11]. However, its response time is in the order of a few tens of picoseconds. The optical nonlinearities induced in an electroabsorption modulator (EAM) have a faster response than those induced in an SOA and a higher efficiency than those in silica fiber. Under typical conditions, the absorption recovery time is less than 10 ps and the required pulse power is less than one hundred milliwatts. For wavelength conversion of randomly coded signal, recovery time is required to be less than the bit-period to reduce patterning effect, which is the fluctuation of output power depending on the logical pattern of the input signal.

This paper describes all-optical wavelength conversion using an EAM in conjunction with delayed interferometer (DI). In Section II, a general explanation of optical nonlinearity induced in an EAM is introduced. In the former parts of Section III, operation principle of an EAM-DI wavelength converter is described, and the quantitative analyses of the optical nonlinearities of the EAM are shown. In the latter parts of Section III, experimental results of wavelength conversion by the EAM-DI wavelength converter are described. All-optical wavelength conversion at a bit-rate as high as 100 Gb/s was demonstrated by the EAM-DI. Its operable wavelength bandwidth at 40 Gb/s was verified to cover almost full C-band range.

II. PHYSICAL MECHANISM OF OPTICAL NONLINEARITY INDUCED IN EAM

A reverse-biased EAM absorbs input light based on Franz-Keldysh (FK) effect if the absorption layer is bulk, or on the Quantum Confinement Stark effect (QCSE) if the absorption layer is multiple quantum well (MQW). When an optical signal (control signal) is injected into a reverse biased EAM, in which another optical signal (probe signal) copropagates, carrier pairs are generated due to absorption of the control signal and are swept out by external electric field. The photo-generated carriers induce (a) screening of the external electric field and (b) band filling. For the MQW absorption layer, if the exciton binding energy is comparable to the thermal energy at

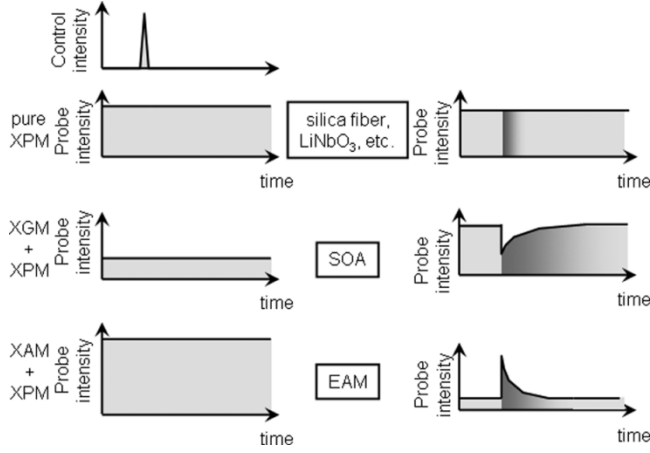


Fig. 1. Schematic illustration of variety of optical nonlinearity.

room temperature ($kT \approx 25$ meV), exciton bleaching should be also taken into account. All these three phenomena work to shift the absorption spectrum to shorter wavelength, and the copropagating probe signal experiences cross-absorption modulation (XAM). Since the real part of propagation constant of light (i.e., absorption coefficient) has Kramers-Kronig relationship with its imaginary part (i.e., refractive index), the refractive index spectrum is also changed by the control signal resulting in cross-phase modulation (XPM) to the probe light.

Fig. 1 illustrates three types of optical nonlinearity including both intensity modulation and phase modulation (expressed as the graduation in the figure) that can be achieved by various devices. As briefly discussed in the former section, pure XPM that is induced in silica fiber or other dielectric devices changes the phase of the probe signal with ultrafast recovery without intensity modulation. In the case of SOA, the probe signal is not only amplified but is also cross-modulated by the control signal both in intensity (cross-gain modulation, XGM) and in phase (XPM) followed by relatively long gain recovery process. In the case of EAM, the probe signal is absorbed and suffers loss, which is one of the problems for the EAM-based wavelength converters, and it is cross-modulated by both XAM and XPM with relatively fast absorption recovery process. Both XAM and XPM induced in an EAM can be utilized for wavelength conversion. There are several previous reports concerning to pure XAM type EAM-based wavelength converter [12]–[19]. In this paper, an advanced type EAM-based wavelength converter, in which XPM is utilized by DI, is investigated.

III. ALL-OPTICAL WAVELENGTH CONVERSION BY EAM WITH DI

A. Operation Principle

When an intense optical pulse (hereafter referred to as “control pulse”) is injected into an EAM, copropagating continuous wave (CW) light (hereafter referred to as “probe light”) experiences both XAM and XPM. We assumed that the absorption of the EAM recovers exponentially with single time constant of τ as expressed by the following:

$$I(t) = I_0 + \Delta I_0 \exp\left(-\frac{t}{\tau}\right) \quad (1)$$

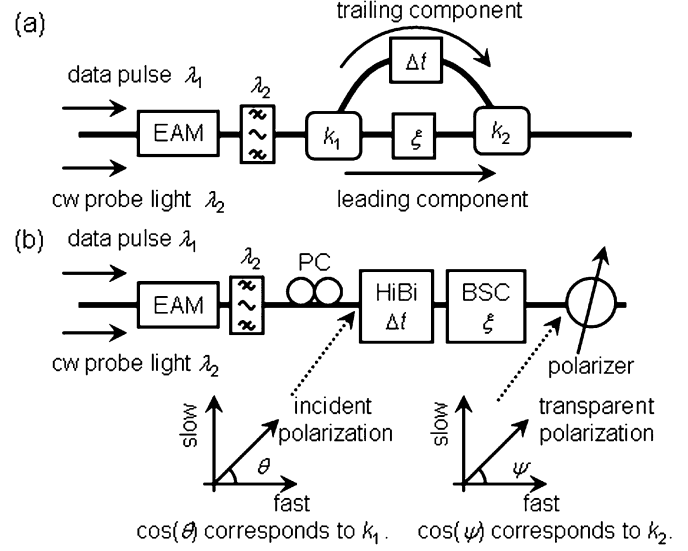


Fig. 2. Configurations of DI. (a) Route discriminated. (b) Polarization discriminated.

where $I(t)$ is the intensity of the output probe light from EAM as a function of time of t , I_0 is the intensity of the output probe light without control pulse injection, ΔI_0 is the initial intensity increment of the output probe light due to XAM, and the control pulse is injected at the time $t = 0$. We also assumed that the phase shift due to XPM also recovers with the same time constant of τ , as expressed by the following:

$$\Delta\phi(t) = \Delta\phi_0 \exp\left(-\frac{t}{\tau}\right) \quad (2)$$

where $\Delta\phi(t)$ is the phase shift as a function of time and $\Delta\phi_0$ is the initial phase shift due to XPM. Not only XAM but also XPM can be utilized for wavelength conversion with interferometric configuration such as Mach-Zehnder interferometer (MZI) or Michelson interferometer. We employed DI configuration [see Fig. 2(a)] due to its structural simplicity and potential high-speed operability to investigate wavelength conversion by an EAM [20]–[23]. The probe light output from the EAM is input into two arms of asymmetric MZI, which is DI that gives differential time delay of Δt between the two components, i.e., the leading component and the trailing component. When no control pulse is input, the initial phase offset between the two components is adjusted by the phase shifter [ξ in Fig. 2(a)] so as that the two components interfere destructively with each other. When a control pulse is injected into the EAM, the intensity of leading component is increased due to XAM. At the same time, the phase shift due to XPM turns the interference constructive to some extent. After the period of Δt , the trailing component started to be also modulated and it interferes with the leading component destructively again. Consequently, the tail of exponentially recovering leading component is cancelled out. In such a way, the wavelength converted pulse with duration of Δt is generated. In the equations, the pulse generation process can be expressed as

$$E_{\text{lead}}(t) = \sqrt{k_1} \sqrt{1 - k_2} \sqrt{I(t)} \times \exp\{j\Delta\phi(t) + \xi\} \text{ for } 0 \leq t \quad (3)$$

$$E_{\text{trail}}(t) = \begin{cases} \sqrt{1-k_1}\sqrt{k_2}\sqrt{I_0} & \text{for } 0 \leq t < \Delta t \\ \sqrt{1-k_1}\sqrt{k_2}\sqrt{I(t-\Delta t)} & \text{for } \Delta t \leq t \end{cases} \quad (4)$$

$$I_{\text{out,DI}}(t) = |E_{\text{lead}} + E_{\text{trail}}|^2 \quad (5)$$

where $E_{\text{lead}}(t)$ and $E_{\text{trail}}(t)$ are the electric fields of the leading component and the trailing component, respectively, as functions of time, k_1 is the power splitting ratio of the input coupler of the DI, k_2 is the power combining ratio at the output coupler of the DI, and ξ is the phase offset applied by the phase shifter. For the wavelength conversion for random sequence pulse, ξ and either k_1 or k_2 should be precisely adjusted because the leading component is partly recovered after the period of Δt and its intensity and the phase do not exactly match to those of the trailing component. In practice, a polarization discriminating DI [PD-DI, see Fig. 2(b)] is more convenient due that the splitting (k_1) and the combining (k_2) ratios of the couplers can be easily tuned by controlling the polarization directions. In addition, PD-DI is superior in stability, since both the leading and the trailing components pass the identical path and experience almost the same environmental fluctuations. Instead of the asymmetric MZI, a highly birefringent device (HiBi), such as a polarization maintaining fiber (PMF), is used to separate the output probe light into the leading polarization component and the trailing polarization component. The phase offset between the two polarization components of ξ can be precisely controlled by a Babinet–Soleil type phase compensator (BSC).

B. Characterization of XAM and XPM in EAM

In the wavelength conversion experiment, a 100 μm -long EAM having MQW absorption layer, of which the photoluminescence peak wavelength at room temperature was 1480 nm, was used. The absorption edge of the EAM was blueshifted by about 30 nm with reference to a typical EAM that was designed as a C-band external intensity modulator, so as that a high bias voltage can be applied to accelerate the absorption recovery without inducing an excessive loss to the probe light. We have quantitatively analyzed the XAM and XPM characteristics of the EAM employing PD-DI configuration, which is the same as that for the wavelength converter but using a control pulse with repetition rate of 10 GHz [24], with the following conditions: the EAM temperature was 0 °C, the wavelength of the control pulse was 1545 nm and its pulse width was 3 ps, and the wavelength of the CW probe light was 1555 nm and its launched power was +3.2 dBm. The static absorption characteristics of the EAM at 0 °C are shown in Fig. 3. The absorption recovery time τ decreases with the increase of bias voltage, V_b , due to acceleration of carrier sweep-out as plotted in Fig. 4, and the absorption recovery time was less than 10 ps at bias voltage higher than -4 V. XAM and XPM characteristics, which were derived based on (1)–(5), are shown in Figs. 5 and 6, respectively. At -4.5 -V bias with 1.5-pJ control pulse injection, the normalized initial increment of output probe light, $\Delta I_0/I_0$, which corresponds to the extinction ratio wavelength converted pulse by pure XAM was 1.9 (2.8 dB). Referring to the static extinction characteristics shown in Fig. 3, the maximum extinction ratio by pure XAM from $V_b = 0$ V to $V_b = -4.5$ V is 3.9 dB. In the EAM-DI configuration, much higher extinction

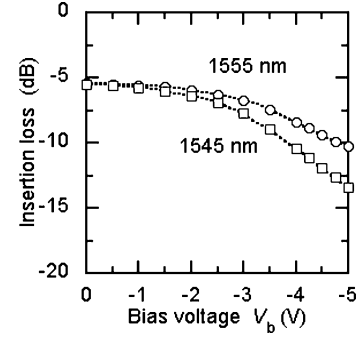


Fig. 3. Static absorption characteristics at 0°.

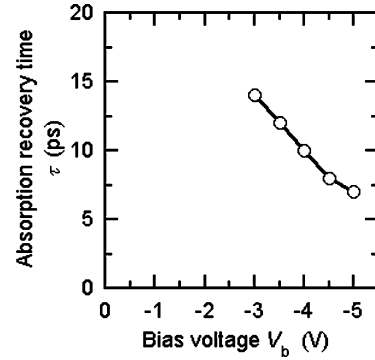


Fig. 4. Absorption recovery time versus bias voltage.

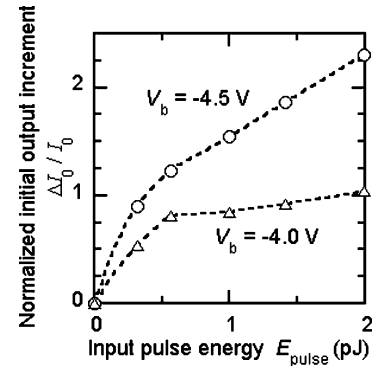


Fig. 5. XAM characteristics.

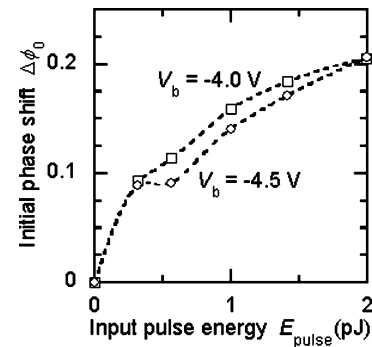


Fig. 6. XPM characteristics.

ratio can be achieved thanks to the combination of destructive interference and phase shift induced by XPM operating DI in push-pull manner, as described in the following section. The initial phase shift $\Delta\phi_0$ was 0.18π at $V_b = -4.5$ V with 1.5-pJ

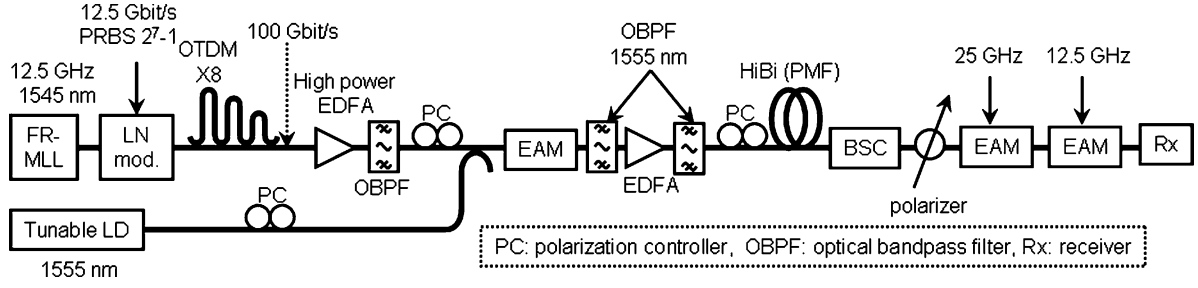


Fig. 7. Experimental setup for 100 Gb/s wavelength conversion.

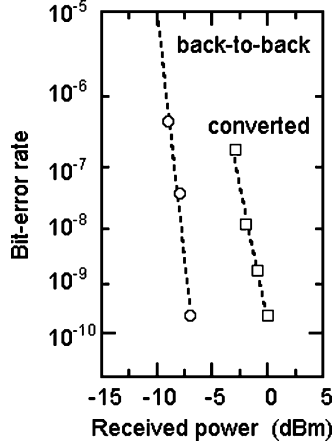


Fig. 8. BER measurement results of 100 Gb/s wavelength conversion from 1545 nm to 1555 nm.

pulse energy, which is not as large as that induced in an SOA, but is large enough to switch DI into constructive.

C. High Bit-Rate Wavelength Conversion Experiment

Fig. 7 depicts schematic experimental setup for 100 Gb/s wavelength conversion by EAM PD-DI. As for the data pulse, 3-ps wide optical pulse with repetition rate of 12.5 Gb/s, which was generated by a fiber-ring mode-locked laser (FR-MLL), was modulated by a LiNbO₃ modulator (LN mod) with a pseudorandom bit sequence (PRBS) of $2^7 - 1$, and the pulse was optically time-division multiplexed (OTDM) up to 100 Gb/s. The respective launched powers were +18.8 dBm for data pulse and +4.8 dBm for CW probe light. The EAM was cooled at 0 °C and V_b was set at -4.3 V. A PMF with polarization mode dispersion (PMD) of 3.2 ps was employed as a HiBi in the PD-DI. The bit-error rate (BER) was measured after optical demultiplexing by cascaded two EAMs from 100 to 25 Gb/s and succeedingly to 12.5 GHz. Fig. 8 shows the BER measurement results. Error-free operation was confirmed, and the power penalty at $\text{BER} = 10^{-9}$ was 6.6 dB. The eye-diagrams of the input signal, wavelength converted signal and demultiplexed wavelength converted signal are shown in Fig. 9. The extinction ratio exceeding 10 dB was obtained, which was much higher than that obtained by pure XAM. A slight patterning effect on the mark level and the intersymbol interference are discernible, to which the power penalty can be attributed. Further blueshift of the absorption edge of the EAM, to which a higher bias voltage can be applied, may be a good way to realize faster absorption recovery

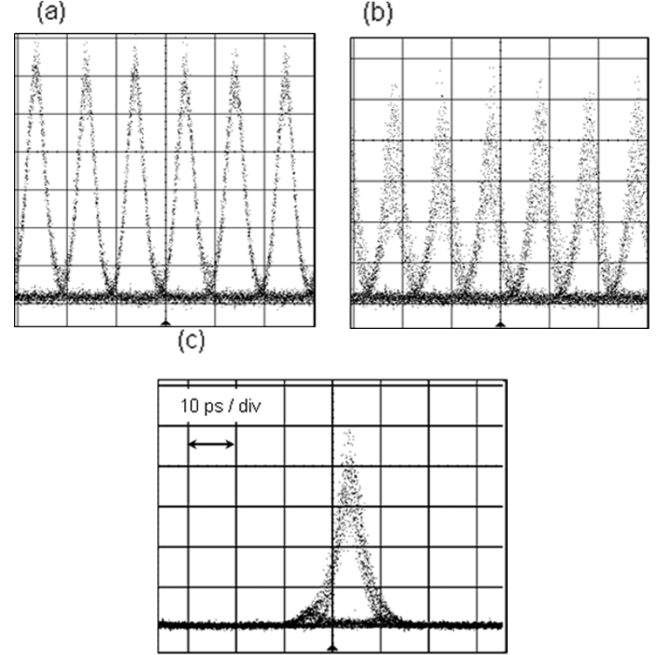


Fig. 9. Eye-diagrams of (a) input data signal, (b) wavelength converted signal, and (c) demultiplexed wavelength converted signal, respectively. Horizontal scales are in 10 ps/div.

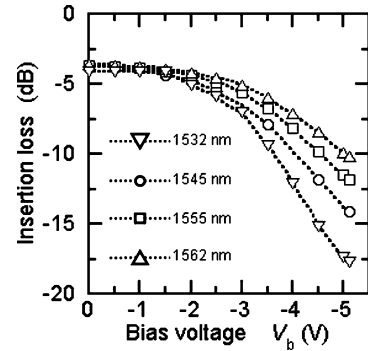


Fig. 10. Static absorption characteristics of the EAM at 10 °C.

without sacrificing absorption loss resulting in further improvement of 100 Gb/s wavelength conversion performances.

D. Wavelength Conversion Experiment Covering C-Band

As the EAM absorption characteristics has a large dependence on the wavelength as shown in Fig. 10 (note that the temperature of EAM was 10°C in this case), we have tested the operable wavelength bandwidth of EAM PD-DI wavelength con-

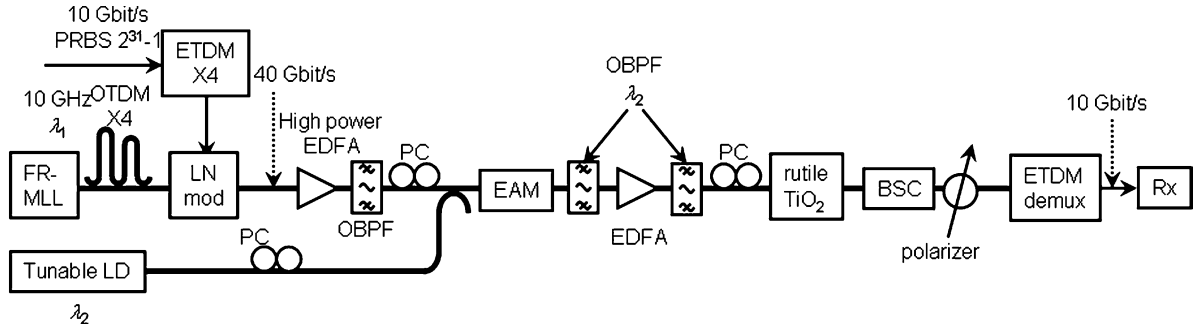


Fig. 11. Experimental setup for wide-band wavelength conversion.

verter at 40 Gb/s. Fig. 11 shows schematic experimental setup. The 40 GHz optical pulse was generated by 10 GHz FR-MLL and OTDM, and successingly the pulse was encoded by 40 Gb/s data signal, which was electrically multiplexed from 10 Gb/s PRBS data of $2^{31} - 1$, using a LiNbO₃ modulator. The shortest wavelength of the data pulse was 1545 nm, which was limited by the gain bandwidth of the high-power EDFA that was used to amplify the data pulse, and the longest wavelength was 1562 nm, which was the longest operating wavelength of the FR-MLL. The wavelength of the probe light was varied between 1532 and 1562 nm. The input optical power was +15 dBm for data pulse and +1.2 dBm for CW probe light, respectively. The EAM was cooled at 10°C and biased at -5.1 V. A rutile type titanium oxide crystal module equipped with input and output fibers having 3.1 ps PMD was employed as HiBi in the PD-DI. An electric demultiplexer was used for extracting four 10 Gb/s tributaries in order to minimize the wavelength dependence of the receiver. BER measurement results for the case of wavelength conversion from 1562 to 1532 nm are shown in Fig. 12 as an example. The power penalty at BER = 10^{-9} was ranging from 0.7 to 1.4 dB in this case. Table I summarizes the tested combinations of wavelengths and the measured power penalties. Error-free operation was confirmed for every combination of wavelengths, and the power penalty was less than 2.5 dB. Note that operation conditions, such as the input optical powers, and the bias voltage and the temperature of EAM, were kept constant for all the cases. The data signal with longer wavelength is less absorbed and penetrates further along to the EA waveguide than the shorter wavelength signal does. As a consequence, the total optical nonlinearity integrated along the propagation direction may be less wavelength-sensitive than the static absorption is, which is shown in Fig. 10. This may be a possible explanation for why such a wide operation bandwidth is attainable for EAM-DI wavelength converter, although more details are still under investigation.

These results indicate that the EAM PD-DI wavelength converter can cover almost full range of C-band without changing operation conditions.

IV. CONCLUSION

All-optical wavelength conversion using an electro-absorption modulator (EAM) in conjunction with delayed-interferometer (DI) was investigated. The optical nonlinearities induced in an EAM were quantitatively characterized by experiment. High bit-rate wavelength conversion at 100 Gb/s was demonstrated by

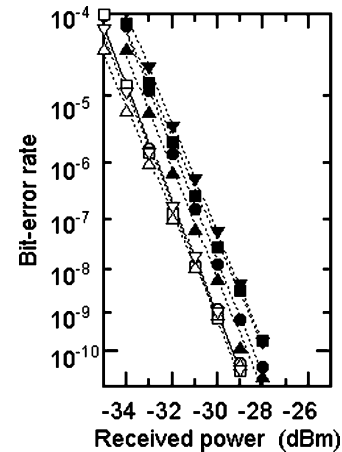


Fig. 12. BER measurement results for 40 Gb/s wavelength conversion from 1562 to 1532 nm. Open symbols are for back-to-back and solid symbols for converted pulse, where the same plot symbol stands for the correspondent tributaries.

TABLE I
TESTED COMBINATIONS OF WAVELENGTHS AND RESULTING POWER PENALTIES

data pulse wavelength	probe light wavelength	measured power penalty
1562nm	1532nm	0.7~1.4dB
1562nm	1555nm	1.9~2.5dB
1555nm	1545nm	0.5~1.1dB
1545nm	1532nm	0.4~1.0dB
1545nm	1555nm	1.4~1.9dB
1545nm	1562nm	0.7~2.1dB

an EAM-DI wavelength converter with required optical pulse energy of 1.5 pJ. It was verified that the operable bandwidth of the EAM-DI wavelength converter at 40 Gb/s covered almost full C-band range without tuning operation conditions. Very recently, we have started theoretical analysis how to reduce the required power of EAM-DI wavelength converter. Preliminary results showed that the required power would be reduced by 4.7 dB employing 37 μ m-long EAM with larger absorption coefficient than that of blue-shifter EAM used in the present study without degrading wavelength conversion performance, although the investigation is not completed yet. Details of the analysis will be reported elsewhere in the future, but relying on the results, further optimization of the EAM design possibly realize ultrafast wavelength conversion with required pulse energy ranging 500 fJ/pulse or less, which is less than one order of magnitude larger than that required for typical SOA-based wavelength converters. These results indicate that the EAM-based

wavelength converter is capable for ultrafast wide-band wavelength conversion.

ACKNOWLEDGMENT

The authors would like to thank T. Asami and M. Suzuki in for their continuous encouragements.

REFERENCES

- [1] B.-E. Olsson, P. Ohlen, L. Rau, and D. J. Blumenthal, "A simple and robust 40-Gb/s wavelength converter using fiber cross-phase modulation and optical filtering," *IEEE Photon. Technol. Lett.*, vol. 12, no. 7, pp. 846–848, 2000.
- [2] Y. Jianjun and P. Jeppesen, "80-Gb/s wavelength conversion based on cross-phase modulation in a high-nonlinearity dispersion-shifted fiber and optical filtering," *IEEE Photon. Technol. Lett.*, vol. 13, no. 8, pp. 833–835, Aug. 2001.
- [3] V. E. Perlin and H. G. Winful, "All-fiber wavelength conversion using cross-phase modulation and Bragg gratings," *IEEE Photon. Technol. Lett.*, vol. 14, no. 2, pp. 176–178, Feb. 2002.
- [4] B. C. Sarker, T. Yoshino, and S. P. Majumder, "All-optical wavelength conversion based on cross-phase modulation (XPM) in a single-mode fiber and a Mach-Zehnder interferometer," *IEEE Photon. Technol. Lett.*, vol. 14, no. 3, pp. 340–342, Mar. 2002.
- [5] J. H. Lee, Z. Yusoff, W. Belardi, M. Ibsen, T. M. Monro, and D. J. Richardson, "A tunable WDM wavelength converter based on cross-phase modulation effects in normal dispersion holey fiber," *IEEE Photon. Technol. Lett.*, vol. 15, no. 3, pp. 437–439, Mar. 2003.
- [6] A. W. O'Neill and R. P. Webb, "All-optical loop mirror switch employing an asymmetric amplifier/attenuator combination," *Electron. Lett.*, vol. 26, no. 24, pp. 2008–2009, 1990.
- [7] S. Nakamura, K. Tajima, and N. Hamano, "High-speed all-optical switching experiment in Mach-Zehnder configuration using GaAs waveguide," *Appl. Phys. Lett.*, vol. 62, no. 9, pp. 925–927, 1993.
- [8] K. Tajima, S. Nakamura, and Y. Sugimoto, "Ultrafast polarization-discriminating Mach-Zehnder all optical switch," *Appl. Phys. Lett.*, vol. 67, no. 25, pp. 3709–3711, 1995.
- [9] K. E. Stubkjaer, "Semiconductor optical amplifier-based all-optical gates for high-speed optical processing," *IEEE J. Select. Topics Quantum Electron.*, vol. 6, no. 6, pp. 1428–1435, Nov./Dec. 2000.
- [10] D. Wolfson, T. Fjelde, A. Kloch, B. Dagens, C. Janz, F. Poinet, I. Guillemot, F. Gaborit, A. Coquelin, and M. Renaud, "All-optical wavelength conversion scheme in SOA-based interferometric devices," *Electron. Lett.*, vol. 36, no. 21, p. 1794, 2000.
- [11] M. Tsurusawa, K. Nishimura, and M. Usami, "Novel scheme for reducing the pattern effect in 40 Gb/s SOA based all-optical switch utilizing transparent CW assist light," *Jpn. J. Appl. Phys.*, pt. 1, vol. 41, no. 2B, pp. 1199–1202, 2002.
- [12] N. Edagawa, M. Suzuki, and S. Yamamoto, "Novel wavelength converter using an electroabsorption modulator," *IEICE Trans. Electron.*, vol. 81, no. 8, pp. 1251–1257, 1998.
- [13] P. S. Chou, D. Mahgerefteh, and J. Goldhar, "All-optical 2R regeneration and wavelength conversion at 20 Gb/s using an electroabsorption modulator," *IEEE Photon. Technol. Lett.*, vol. 11, no. 12, pp. 1662–1664, Dec. 1999.
- [14] T. Otani, T. Miyazaki, and S. Yamamoto, "Optical 3R regenerator using wavelength converters based on electroabsorption modulator for all-optical network applications," *IEEE Photon. Technol. Lett.*, vol. 12, no. 4, pp. 431–433, Apr. 2000.
- [15] S. Højfeldt, S. Bischoff, and J. Mørk, "All-optical wavelength conversion and signal regeneration using an electroabsorption modulator," *J. Lightw. Technol.*, vol. 18, no. 8, pp. 1121–1127, Aug. 2000.
- [16] E. S. Awad, P. S. Cho, C. Richardson, N. Moulton, and J. Goldhar, "Optical 3R regeneration using a single EAM for all-optical timing extraction with simultaneous reshaping and wavelength conversion," *IEEE Photon. Technol. Lett.*, vol. 14, no. 9, pp. 1378–1380, Sep. 2002.
- [17] K. K. Chou and C. Shu, "All-optical wavelength conversion with multicasting at 6×10 bit/s using electroabsorption modulator," *Electron. Lett.*, vol. 39, no. 19, pp. 1395–1397, 2003.
- [18] H.-F. Chou, Y.-J. Chiu, A. Keating, J. E. Bowers, and D. J. Blumenthal, "Photocurrent assisted wavelength (PAW) conversion with electrical monitoring capability using a traveling-wave electroabsorption modulator," *IEEE Photon. Technol. Lett.*, vol. 16, no. 2, pp. 530–532, Feb. 2004.
- [19] N. E. Dahdah, G. Aubin, J.-C. Harmand, A. Ramdane, A. Shen, F. Devaux, A. Garreau, and B.-E. Benkelfat, "Ultrafast InGaAs/InGaAlAs multiple-quantum-well electro-absorption modulator for wavelength conversion at high bit rates," *Appl. Phys. Lett.*, vol. 84, no. 21, pp. 4268–4270, 2004.
- [20] Y. Ueno, S. Nakamura, K. Tajima, and S. Kitamura, "3.8-THz wavelength conversion of picosecond pulses using a semiconductor delayed-interference signal-wavelength converter (DISC)," *IEEE Photon. Technol. Lett.*, vol. 10, no. 3, pp. 346–348, 1998.
- [21] J. Leuthold, B. Mikkelsen, G. Raybon, C. H. Joyner, J. L. Pleumeekers, B. I. Miller, K. Dreyer, and R. Behringer, "All-optical wavelength conversion between 10 and 100 Gb/s with SOA delayed-interference configuration," *Opt. Quantum Electron.*, vol. 33, no. 7/10, pp. 939–952, 2001.
- [22] K. Nishimura, M. Tsurusawa, and M. Usami, "First demonstration of 40 Gbps wavelength conversion with no pattern effect utilizing cross-phase modulation in an electroabsorption waveguide," *Jpn. J. Appl. Phys.*, pt. 1, vol. 41, no. 2B, pp. 1178–1181, 2002.
- [23] K. Nishimura, R. Inohara, M. Tsurusawa, and M. Usami, "80 Gbit/s wavelength conversion using MQW electroabsorption modulator in delayed-interferometric configuration," *Electron. Lett.*, vol. 39, no. 10, pp. 792–794, 2003.
- [24] R. Inohara, K. Nishimura, M. Tsurusawa, and M. Usami, "Experimental analysis of cross phase modulation and cross gain modulation in SOA injecting CW assist light," *IEEE Photon. Technol. Lett.*, vol. 15, no. 9, pp. 1192–1194, 2003.



Kōsuke Nishimura received the B.S. and M.S. degrees in electrical engineering from Tokyo Institute of Technology, Tokyo, Japan, in 1986 and 1988, respectively.

Since 1988, he has been with KDD R&D Laboratories, Saitama, Japan, where he has studying visible light emission materials including II–VI compound semiconductor and porous silicon. Currently, he is engaged in the research and development of all-optical regeneration, wavelength conversion technologies, and all-optical functional devices for future photonic network.

Mr. Nishimura is a member of the Japan Society of Applied Physics and the Institute of Electronics, Information, and Communications Engineers of Japan.



Ryo Inohara received the B.E. and M.E. degrees in communications engineering from Osaka University, Osaka, Japan, in 1999 and 2001, respectively.

He joined KDDI R&D Laboratories Inc., Saitama, Japan, in 2001. Currently, he is engaged in the research and development of all-optical regeneration and wavelength conversion technologies using semiconductor optical amplifier.

Mr. Inohara is a member of the Japan Society of Applied Physics and the Institute of Electronics, Information, and Communications Engineers of Japan.



Masashi Usami (M'00) received the B.S., M.S., and Ph.D. degrees in electronic and communication engineering from Waseda University, Tokyo, Japan, in 1981, 1983, and 1992, respectively.

Since 1985, he has been with KDD R&D Laboratories, Saitama, Japan, where he has been studying InGaAsP–InP DFB lasers and InGaAs–GaAs high-power lasers. From 1989 to 1990, he was a Fellow of the Center for Advanced Educational Services, Massachusetts Institute of Technology, Cambridge, where he studied bistability of resonant tunneling diode. Currently, he is engaged in the research and development of all-optical regeneration, wavelength conversion technologies, and all-optical functional devices for future photonic network.

Dr. Usami is a member of the Japan Society of Applied Physics and the Institute of Electronics, Information, and Communication Engineers of Japan.



Shigeyuki Akiba (M'88–SM'89–F'02) received the B.S., M.S., and Ph. D. degrees in electronics engineering from Tokyo Institute of Technology, Tokyo, Japan, in 1974, 1976, and 1984, respectively.

He joined KDDI R&D Laboratories, Saitama, Japan, in 1976, where he worked in the area of optoelectronic devices for fiber-optic communications. From 1980 to 1981, he was a Fellow of the Center for Advanced Educational Services (CAES), Massachusetts Institute of Technology, Cambridge, where he studied integrated optics and semiconductor laser mode-locking. From 1988 to 1990, he was with the R&D Department, Intelsat, Washington, D.C. Since 1990, he had been involved in the R&D program toward optical amplifier undersea cable systems, especially ultra-high capacity advanced WDM systems, which include TPC-5 Cable Network, Japan-US, PC-1, and TAT-14. In 2000, he became President and Chief Executive Officer (CEO) of KDD R&D Laboratories, Inc. He is currently President and CEO of KDDI Submarine Cable Systems, Inc., Tokyo, Japan. He has served as a Co-Chair of Program Committee (2000) and Organizing Committee (2002) of Optoelectronics and Communications Conference (OECC).

Dr. Akiba is a member of Institute of Electronics, Information, and Communications Engineers (IEICE) of Japan. He is a recipient of Young Engineers Award of IEICE and JSAP (Japan Society of Applied Physics) in 1979, Maejima Award in 1995, Distinguished Paper Award of IEICE in 1996, and Achievement Award of IEICE in 1997.